Piezoelectric Energy Harvesting Research For Self-Sustainable Wind Turbine Monitoring Systems

Miles Smith, Alexander Gliese, HyunJun Jung, and Soobum Lee

Department of Mechanical Engineering

Background

Current Problem:

- Wind energy is still costly due to high maintenance cost, notably the cost to replace sensors (used to detect cracks and damage within a wind turbine) is very high.
- Current wired sensor networks:
  - Increase the risk of lightning attraction.
  - Low durability
- Current wireless sensor networks:
  - Dependent on batteries that are difficult to replace.

The Piezoelectric Solution:

- A piezoelectric transducer (PZT) could convert mechanical stress into electric potential (voltage), which could then be turned into a power.
- Using ambient energy from the rotation of a wind turbine, usable power can be created from a PZT.

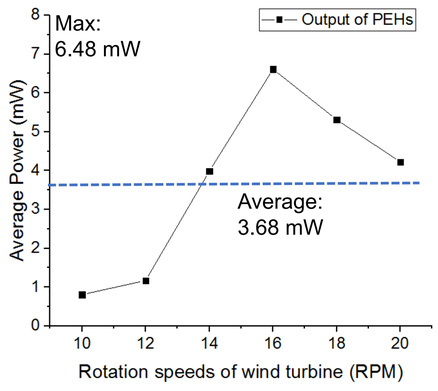
Research Question

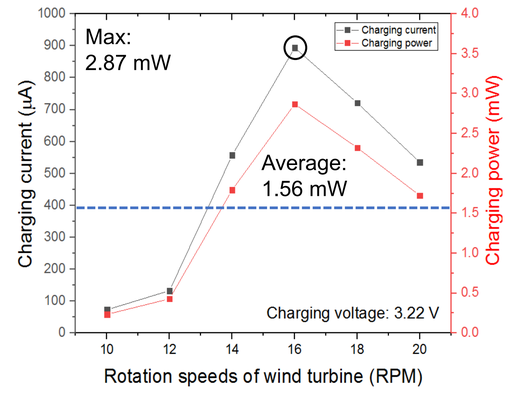
- Can a piezoelectric energy harvesting system be used to replace batteries for sensors to reduce maintenance costs?

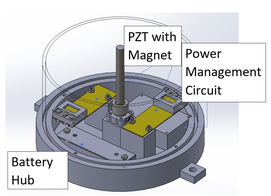
Methods

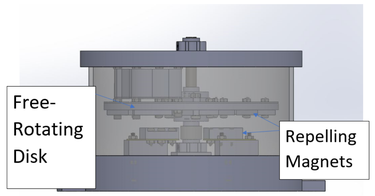
- A scaled wind turbine system was used to determine the power output of the system during rotational speeds of a wind turbine (10 to 20 RPM).
- Conditions of the piezoelectric energy system optimized to maximize power output including:
  - Gap between the magnets on the rotating disk and on the PZT,
  - Weight of the unbalanced disk
  - Number of magnets on the disk
  - Electrical Resistance

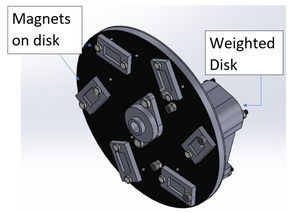
Results











- Resistance = 30 kΩ.
- Unaltered power output based on Ohm's Law ($P = V^2/R$, where P is power, V is voltage and R is resistance).
- The current power management circuit (PMC) operates at an approximate efficiency of 47.6%, which results in a reduction of power output.

Discussion

- Target output power was 3 mW.
  - Achieved by the combination of 2 PZTs, before PMC was applied.
- Suggests the potential for battery replacement in sensors.
- Inconsistent power output at different rotational speeds.
  - Power output peaked at 16 RPM.
  - Significant decrease under 14 RPM.

Future Directions

- Energy harvesting system should be improved to have a more consistent power output at different rotational speeds.
- Size and cost both could be significantly reduced.
- Reducing the system from 2 PZTs to 1 could help to significantly reduce cost and size if power level could be maintained.

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